

Term Deposit Subscription Prediction – Executive Summary & Technical Report

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1. Executive Summary

1.1 Purpose

The purpose of this project is to develop a predictive model that estimates whether a bank client will subscribe to a term deposit during a direct marketing campaign. The goal is to support the marketing and sales teams by identifying high-propensity customers, improving the efficiency of outbound phone calls, and increasing conversion rates for long-term savings products.

1.2 Approach

A complete data science workflow was followed:

- Exploratory Data Analysis (EDA)
- Data preprocessing
- Predictive modeling (multiple algorithms)
- SHAP-based model explainability
- Business recommendations and management-level strategy
- Cloud deployment proposal and implementation timeline

Four machine learning models were trained and evaluated using metrics suitable for imbalanced binary classification:

- Logistic Regression (baseline, interpretable)
- Random Forest
- Gradient Boosting
- XGBoost (final recommended model)

1.3 Key Findings

- The dataset contains **41,188 records** with a clear class imbalance:
 - **11.3%** of clients subscribed to a term deposit ("yes")
 - **88.7%** did not ("no")
- **Model performance (positive class = "yes"):**
 - **Logistic Regression**
 - ROC-AUC = **0.803**
 - Recall = **0.647**
 - F1-score = **0.465**

- **Random Forest**
 - ROC-AUC = **0.781**
 - Recall = **0.269**
 - F1-score = **0.364**
- **Gradient Boosting**
 - ROC-AUC = **0.810** (highest ROC-AUC)
 - Recall = **0.241**
 - F1-score = **0.353**
- **XGBoost**
 - ROC-AUC = **0.809**
 - Recall = **0.652** (highest recall)
 - F1-score = **0.485** (best F1)
- **Model selection conclusion:**

XGBoost provides the **best business value** due to:

 - highest recall on the positive class (captures the most true subscribers),
 - best F1-score (balanced precision–recall),
 - competitive ROC-AUC comparable to Gradient Boosting.
- **SHAP explainability** shows that the strongest predictors of subscription include:
 - **age**
 - **euribor3m** (3-month Euribor rate)
 - **campaign** (number of contacts in current campaign)
 - **nr.employed**
 - **emp.var.rate**
 - **cons.conf.idx**
 - **cons.price.idx**
 - **pdays**
 - **housing loan status (housing_no / housing_yes)**
 - plus categories related to **job, marital status, education, and previous campaign success (poutcome_success)**

1.4 High-Level Recommendations

- **Deploy XGBoost as the production model**, as it achieves the best combination of recall, F1-score, and ranking capability.
- Use the model's probability scores to **prioritize outbound calls** toward high-propensity customers rather than treating all leads equally.
- **Limit repeated calls** to customers with very low predicted propensity to reduce operational costs and avoid customer fatigue.
- **Tailor communication strategies** and product positioning based on SHAP-derived segments (age, job, macroeconomic sensitivity, past campaign response).

- Incorporate **SHAP explanations into internal dashboards** to increase transparency and ensure alignment with compliance and risk teams.

1.5 Next Steps

- Deploy the XGBoost model as a scoring service in a cloud environment (Azure or AWS).
 - Integrate the model with the bank's CRM and dialing systems to drive call prioritization.
 - Implement MLOps practices for:
 - monitoring model performance,
 - detecting data and concept drift,
 - retraining the model (e.g., quarterly).
 - Design **management dashboards** to monitor campaign performance, model lift, and key KPIs.
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2. Business Context and Problem Statement

The Portuguese bank conducts direct marketing campaigns using phone calls to promote term deposits (long-term savings accounts). These campaigns often involve multiple contact attempts to reach each client, which:

- increases operational costs,
- risks irritating customers, and
- dilutes the effectiveness of each marketing cycle.

Currently, all eligible customers may be contacted with similar priority, largely ignoring their individual probability of accepting the offer. This leads to:

- a **low overall conversion rate**, and
- **inefficient use of call center capacity** and agent time.

2.1 Business Question

Can we predict which clients are more likely to subscribe to a term deposit before (or at the very beginning of) the call, so that the marketing team can focus on the most promising prospects?

2.2 Objective

The main objective is to build a **classification model** that estimates the probability that a given client will subscribe to a term deposit. This probability score will:

- enable **ranking and prioritization** of customers within each campaign, and
- form the core of a **data-driven call strategy** that maximizes conversions per call.

2.3 Out of Scope

The following are explicitly out of scope:

- Detailed call center staffing optimization and workforce scheduling.
 - Designing or scripting the full call dialogue (beyond strategic recommendations).
 - Optimization of marketing channels other than phone calls (e.g., email, SMS, digital ads).
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3. Scope and Methodology

A standard **CRISP-DM style** methodology was applied, from understanding the business to deploying actionable insights.

3.1 Exploratory Data Analysis (EDA)

EDA focused on:

- understanding distributions and ranges for each feature,
- studying target imbalance and segment-wise subscription rates,
- identifying and interpreting “unknown” categories,
- exploring relationships between features and the target variable **y**,
- assessing macroeconomic trends and their correlation with term deposit subscriptions.

3.2 Data Preprocessing

Key preprocessing steps:

- Removal of features causing **target leakage** (e.g., duration).
- Categorical encoding using **one-hot encoding**.
- **Standardization** of numerical features for models sensitive to scale (e.g., Logistic Regression).
- **Stratified train/test split** to preserve the target’s imbalance structure.

3.3 Predictive Modeling

Four models were trained and compared:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost

Models were evaluated with metrics appropriate for imbalanced data: ROC-AUC, precision, recall, F1-score, and confusion matrix analysis.

3.4 Feature Interpretation

Model interpretability was achieved through:

- global **feature importance** from XGBoost,
- **SHAP TreeExplainer** to quantify feature contributions, both globally and at individual customer level.

3.5 Business Recommendations

Findings from the modeling phase were translated into:

- a set of **operational recommendations** (call prioritization, contact strategy), and
- a broader **management-level strategic plan** to improve long-term savings product adoption.

3.6 Cloud Deployment Architecture

A high-level deployment proposal, based on **Microsoft Azure**, was designed to:

- store data,
 - train and deploy the model,
 - expose it as an API for CRM integration, and
 - monitor both technical and business performance.
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4. Exploratory Data Analysis – Key Findings

4.1 Dataset Overview

- **41,188 observations**
- **21 variables:**
 - 20 input features
 - 1 binary target (y: "yes" / "no")
- **Target imbalance:**
 - approx. **4,640** "yes" (11.3%)
 - approx. **36,548** "no" (88.7%)

4.2 Data Quality and "Unknown" Values

Several categorical variables include an explicit **"unknown"** category, which is preserved as a meaningful signal:

- job: ~0.8% "unknown"
- education: ~4.2% "unknown"
- default: ~20.9% "unknown"
- housing: ~2.4% "unknown"
- loan: ~2.4% "unknown"

Rather than imputing or dropping these, they were retained as independent categories, recognizing that "missing" information may correlate with specific customer profiles.

4.3 Target Leakage and the **duration** Variable

The variable **duration** (last call length in seconds) is highly predictive:

- Longer calls tend to be associated with a higher probability of subscription.

However:

- duration is only known **after the call finishes**, at which point the outcome is already determined.
- Using it to predict before the call would constitute **target leakage**.

Therefore:

- **duration was excluded** from the production model and only considered for exploratory/benchmarking purposes.

4.4 Behavioral and Contextual Patterns

Key patterns identified:

- **Previous campaign outcome** (`poutcome`) is strongly associated with current subscription: customers with a previous successful outcome are much more likely to subscribe again.
 - **Number of contacts** in the current campaign (`campaign`) shows diminishing returns: as the number of calls increases beyond a small threshold, **subscription probability decreases**, indicating customer fatigue.
 - **Macroeconomic indicators** (`euribor3m`, `emp.var.rate`, `cons.conf.idx`, `cons.price.idx`) are closely tied to subscription behavior, suggesting that customers are more receptive to term deposits under certain economic conditions.
 - **Customer profile variables** such as age, job, and education reveal segments with distinct propensity levels, useful for segmentation and tailored marketing.
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5. Data Preprocessing

5.1 Variable Selection

- The **duration** variable was removed to avoid target leakage.
- All other features (demographics, campaign variables, macroeconomic indicators) were retained.

5.2 Handling "Unknown" Categories

- "Unknown" values were treated as valid categories (not imputed):
e.g., `job = "unknown"`, `education = "unknown"`, etc.
- This approach preserves potential signal associated with missing information patterns.

5.3 Encoding Categorical Variables

- Categorical features were transformed via **one-hot encoding**.
- This avoids imposing artificial order relationships and works well for both linear and tree-based models.

5.4 Scaling Numerical Variables

- Numerical features (e.g., age, campaign, pdays, previous, macro variables) were **standardized** when used with Logistic Regression.
- Tree-based models (Random Forest, Gradient Boosting, XGBoost) can operate effectively without scaling but were fed the same transformed pipeline for consistency.

5.5 Train/Test Split and Validation

- The dataset was split into **training and test sets** using a **stratified** split on the target `y`.
 - This preserved the original class imbalance in both subsets.
 - Further robustness can be achieved via **k-fold cross-validation** for hyperparameter tuning (especially for XGBoost).
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6. Predictive Modeling and Evaluation

6.1 Models Considered

1. Logistic Regression

- Simple, interpretable baseline.
- Provides direct insight into direction and strength (log-odds) of each feature.

2. Random Forest

- Bagging-based tree ensemble.
- Robust to noise, can capture nonlinear interactions.

3. Gradient Boosting Classifier

- Boosting-based tree ensemble.
- Often achieves strong performance on structured, tabular data.

4. XGBoost (eXtreme Gradient Boosting)

- Optimized gradient boosting implementation.
- Strong performance, handles imbalance via weighting, and provides rich importance metrics.
- Selected as **final production model**.

6.2 Evaluation Metrics

Given the strong class imbalance, accuracy is not informative. Instead, we focus on:

- **ROC-AUC**: ability to rank positive cases above negative ones.
- **Precision (yes)**: proportion of predicted "yes" that are actually "yes".
- **Recall (yes)**: proportion of real "yes" that are correctly identified.
- **F1-score (yes)**: harmonic mean of precision and recall.
- **Confusion matrix**: supports interpreting trade-offs in false positives and false negatives.

6.3 Model Performance

Logistic Regression

- ROC-AUC = **0.803**
- Recall (yes) = **0.647**
- F1-score (yes) = **0.465**

Random Forest

- ROC-AUC = **0.781**
- Recall (yes) = **0.269**
- F1-score (yes) = **0.364**

Gradient Boosting

- ROC-AUC = **0.810** (highest ROC-AUC)

- Recall (yes) = **0.241**
- F1-score (yes) = **0.353**

XGBoost

- ROC-AUC = **0.809**
- Recall (yes) = **0.652** (highest)
- F1-score (yes) = **0.485** (best)

6.4 Model Selection

While Gradient Boosting slightly outperforms XGBoost on ROC-AUC (0.810 vs 0.809), XGBoost clearly dominates in:

- **Recall** (0.652 vs 0.241), and
- **F1-score** (0.485 vs 0.353).

Given the business goal—**not missing potential subscribers** (high recall)—and the need for a balanced performance, **XGBoost is selected as the final production model** due to:

- Highest recall on the positive class,
- Best F1-score,
- Competitive ROC-AUC,
- Strong feature interpretability via SHAP.

7. Feature Importance and Interpretation

7.1 Key Drivers (XGBoost Feature Importance)

Top features by importance:

1. **age** (~0.133)
2. **euribor3m** (~0.125)
3. **campaign** (~0.076)
4. **nr.employed** (~0.060)
5. **emp.var.rate** (~0.041)
6. **cons.conf.idx** (~0.029)
7. **cons.price.idx** (~0.024)
8. **pdays** (~0.022)
9. **housing_no / housing_yes** (~0.018 each)
10. Additional influential features:
 - **job_admin.**
 - **marital_married, marital_single**
 - **education_high.school, education_university.degree**
 - **poutcome_success**
 - **day_of_week** categories (mon, thu, wed, fri)

7.2 Business Interpretation

- **Economic context** strongly affects client willingness to subscribe:
macro features like **euribor3m**, **nr.employed**, **emp.var.rate**, and **cons.conf.idx** drive perceived risk, opportunity, and preference for fixed income products.
 - **Age** is the top demographic predictor:
older customers tend to exhibit higher preference for secure, long-term savings.
 - **Campaign intensity** (**campaign**, **pdays**) reveals **diminishing returns**:
repeatedly contacting the same customer reduces the probability of success, indicating **customer fatigue**.
 - **Previous positive experience** (**poutcome_success**) significantly boosts the chance of current subscription:
customers who converted in past campaigns form a high-value, warm-lead segment.
 - **Loan status** (particularly housing loans) reflects debt burden and liquidity, influencing willingness to lock funds in a term deposit.
 - **Day-of-week effects** suggest that customers may respond differently depending on when they are contacted, opening an opportunity for **temporal optimization of campaigns**.
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8. Business Recommendations (Operational Level)

8.1 Prioritize High-Propensity Customers

- Use the **XGBoost probability scores** to rank customers by subscription likelihood.
- Focus first on the **top 20–30% of leads** for outbound calling per campaign cycle.
- This maximizes the return on each call and improves conversion per agent-hour.

8.2 Optimize the Number of Contact Attempts

- Define and enforce a **maximum number of calls per customer** (e.g., 2–3 attempts).
- Stop calling customers with persistently low scores and no engagement.
- This reduces costs and avoids degrading the brand through excessive contact.

8.3 Segment and Tailor Communication

- Build segments based on:
 - age groups,
 - job categories,
 - education level,
 - previous campaign outcome,
 - macroeconomic context.
- Tailor messaging accordingly:
 - Younger segments: emphasize **wealth building and long-term growth**.
 - Older segments: emphasize **security, guaranteed returns, and retirement planning**.
 - Previous-success customers: promote **loyalty benefits and exclusive rates**.

8.4 Experiment with A/B Testing

- Run A/B (or A/B/N) tests to compare:
 - current “non-model” strategy vs model-based prioritization,
 - different offer levels or incentives per segment,
 - alternative calling schedules and scripts.
- Use statistically significant results to refine campaign strategy.

8.5 Continuous Monitoring and Model Governance

- Monitor:
 - overall and segment-level conversion rates,
 - model performance (ROC-AUC, recall, F1-score),
 - economic indicators that are strongly tied to model predictions.
 - Retrain the model:
 - on a **quarterly basis**, or
 - whenever significant drift in data distribution or performance is detected.
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9. Strategic Management Plan to Increase Term Deposit Subscriptions

The following recommendations are aimed at **management and executive stakeholders**, translating data science insights into a broader strategy.

9.1 Shift from “Volume of Calls” to “Value of Calls”

Insight:

The model shows that **excessive contact attempts reduce subscription probability**—a symptom of customer fatigue.

Strategy:

- Move away from cold calling to **predictive call prioritization**:
 - Call the **top 20–30% highest-propensity customers first**.
 - Reduce or stop calls to very low-propensity segments.
 - Establish a **maximum call attempts limit** (e.g., 2–3 per cycle).
 - Establish a **maximum call attempt limit** (e.g., 2–3 attempts per campaign cycle).
 - Use predicted probability thresholds to dynamically allocate call center effort.

Expected Impact:

Adding behavioral segmentation powered by predictive scoring allows management to:

- Increase conversion rates **without increasing call volume**
- Reduce operational expenditure (fewer wasted calls)
- Improve customer satisfaction (less fatigue)
- Train agents with **segment-specific scripts**

- Schedule calls around economic trends and customer readiness

Concrete Examples of High- and Low-Propensity Clients (Based on Real Model Insights)

Below are realistic, data-driven examples constructed from your model’s strongest predictors (age, euribor3m, cons.conf.idx, campaign count, previous campaign outcome, pdays, job type, housing loan, etc.).

These examples help management understand *how the model thinks* and why prioritization is necessary.

Example 1 — High-Propensity Client (Strong Candidate for Early Calling)

Client Profile

Feature	Value
age	52
job	"admin."
education	"university.degree"
housing_loan	"no"
poutcome	"success"
previous contacts	1
campaign contacts (current campaign)	1
pdays	30
euribor3m	1.25% (historically favorable period)
consumer confidence index (cons.conf.idx)	relatively high
nr.employed	rising (good economic sentiment)

Why the Model Predicts High Propensity

- Previously subscribed successfully → **very strong positive predictor**
- Low campaign contacts → **no fatigue**
- Favorable macroeconomic indicators → **higher willingness to invest**
- Age range (45–60) → **segment most receptive**
- Stable job + higher education → **higher savings capacity**
- No housing loan → **more liquidity available**

Example Model Score

Predicted Probability of Subscription: ~0.69–0.75

Action

Call early (Top 20%)

Expect significantly higher conversion likelihood.

Example 2 — Low-Propensity Client (Better to Deprioritize or Avoid Calling)

Client Profile

Feature	Value
age	23
job	"blue-collar"
education	"unknown"
housing_loan	"yes"
personal_loan	"yes"
poutcome	"failure"
previous contacts	3
campaign contacts (current campaign)	7 (high fatigue)
pdays	999 (never contacted successfully before)
euribor3m	higher (less attractive for deposits)
consumer confidence index	low (economic uncertainty)

Why the Model Predicts Low Propensity

- Previous outcome = **failure** → negative predictor
- Many past and current contacts → **strong fatigue effect**
- High call count (7) → historically reduces conversion likelihood sharply
- Younger age segment → lower interest in long-term savings
- Two loans → reduced liquidity
- "Unknown" education status → correlates with lower conversion
- Negative macroeconomic context → customers risk-averse

Example Model Score

Predicted Probability of Subscription: ~0.03–0.07

Action

Deprioritize or skip calling

If resources allow: maximum 1 attempt.
Cost of calling outweighs expected return.

Example 3 — Medium-Propensity Client (Worth Calling with Customized Messaging)

Client Profile

Feature	Value
age	38
job	"technician"
education	"high.school"
housing_loan	"no"
personal_loan	"no"
poutcome	"nonexistent"
previous	0
campaign	2
pdays	120
euribor3m	slightly favorable
cons.conf.idx	average

Why the Model Predicts Medium Propensity

- No fatigue but also no prior success
- Reasonable age and stable employment
- No major financial constraints
- Neutral to moderately favorable economic context

Example Model Score

Predicted Probability: ~0.22–0.35

Action

Call after top segment

Tailor script to financial planning & future goals
Ideal for batch #2 or during off-peak call hours

9.2 Leverage Macroeconomic Timing

Insight:

Features like **euribor3m**, **consumer confidence**, and **employment variation** significantly influence subscription behavior.

Strategy:

- Time campaigns based on an **“Economic Opportunity Index”** combining macro indicators.
- Prioritize campaigns when:
 - customer confidence is rising,
 - interest rates make fixed-term products attractive,
 - unemployment is falling.

Expected Impact:

- Campaigns are better aligned with customer readiness to invest.
- Subscription rates increase without necessarily increasing call volume.

9.3 Build Lifetime Value Segments Using SHAP Insights

Insight:

SHAP reveals:

- Age as the strongest demographic predictor.
- Job and education as indicators of financial planning behavior.

Strategy:

Create segment-specific journeys:

Customer Segment	Recommended Messaging	Reason
Young professionals (25–35)	Emphasize <i>wealth building & growth</i>	Responsive to long-term planning
Mid-career (35–50)	Emphasize <i>security + stable returns</i>	Higher liquidity + more conservative
Retirees / pre-retirees	Emphasize <i>protection + guaranteed income</i>	Safety-first decision-making
Previous-success customers	Loyalty offers / exclusive rates	High likelihood to convert again

Expected Impact:

- Higher emotional resonance.
- Increased conversion compared with generic messaging.

9.4 Use Behavioral Economics Principles in Scripts

Small changes in framing significantly affect outcomes.

Recommended Framing Techniques:

- **Loss Aversion:**
"By not locking this rate now, you could miss a period of higher yield."
- **Goal-Based Framing:**
"This term deposit helps ensure you reach your 5–10 year financial goals."
- **Scarcity / Exclusivity (Ethically):**
"We have a limited promotional rate available this quarter."
- **Future Self Projection:**
"Imagine having a guaranteed nest egg in 3–5 years."

Expected Impact:

- Improved persuasion without increasing call volume.
 - Consistent uplift across customer segments.
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9.5 Introduce a Predictive "Next Best Offer" Engine

Not every customer should get the same product pitch.

Strategy:

Use the model + business rules to assign customers to:

- Term Deposit Premium
- Term Deposit Standard
- Short-Term Savings
- Long-Term Savings

Decision logic example:

- **Low propensity:** offer simpler, lower-commitment products.
- **Medium propensity:** offer term deposit with incentive/bonus.
- **High propensity:** confidently recommend the term deposit as primary product.

Expected Impact:

- Reduced rejection rate.
 - Higher overall satisfaction.
 - Better cross-sell and up-sell opportunities.
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9.6 Replace Generic Scripts with AI-Generated Personalized Scripts

Insight:

Generative AI is not required for prediction, but is extremely valuable for **communication**.

Strategy:

Use Generative AI to:

- Automatically generate custom scripts based on:
 - age,
 - job,
 - previous interactions,
 - economic context,
 - SHAP-derived key drivers.

Example script line:

"Based on your financial profile and current interest rates, this term deposit aligns very well with your long-term financial stability goals."

Expected Impact:

- More persuasive, personalized calls.
 - Faster agent preparation times.
 - Increased customer trust and perception of relevance.
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9.7 Optimize the Offer, Not Only the Targeting

Strategy:

- Introduce **early commitment bonuses** for fast decisions.
- Offer **rate guarantees** during volatile economic periods.
- Bundle term deposits with:
 - improved credit conditions,
 - simple insurance benefits,
 - free financial planning sessions.

Expected Impact:

- Converts "maybe" customers into "yes" customers.
 - Increases perceived value of the term deposit product.
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9.8 Implement A/B/N Testing as a Standard Practice

Variables to Test:

- Script variations.
- Incentive levels.
- Call scheduling and day-of-week strategies.
- Segmentation rules and threshold cut-offs.
- "Next best offer" mapping logic.

Expected Impact:

- Continuous, data-driven improvement.
 - Measurable uplift attributable to specific strategic levers.
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9.9 Create KPI Dashboards for Management

Recommended KPIs:

- Conversion rate by segment (age, job, previous outcome).
- Conversion per call attempt.
- Lift vs. baseline (no model strategy).
- Agent performance with vs. without AI-enhanced scripts.
- Monthly economic sentiment vs. term deposit conversions.

Objective:

Enable management to make **evidence-based decisions** rather than relying solely on intuition.

9.10 Establish Quarterly Model Retraining & Governance

Term deposit behavior is highly sensitive to:

- interest rates,
- macroeconomic conditions,
- customer sentiment.

Governance Recommendation:

- Retrain model **every 3 months**, or earlier if strong drift is detected.
- Monitor drift in economic features and target distribution.
- Adjust decision thresholds according to business goals (growth vs. risk).
- Ensure alignment with **regulatory and compliance** frameworks.

Expected Impact:

- Maintains model accuracy and stability.
 - Protects the bank from performance degradation in changing economic regimes.
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10. Cloud Deployment Architecture (Azure Example)

10.1 Core Azure Services

- **Azure Blob Storage**

Stores raw data, processed datasets, and batch-scored outputs.

- **Azure Machine Learning**

Used for:

- experiment tracking,
- model training,
- hyperparameter tuning,
- versioning,
- deployment (real-time or batch endpoints).

- **Azure App Service / Azure Functions**

Exposes the XGBoost model as a REST API for:

- CRM,
- outbound dialer,
- internal applications.

- **Azure SQL Database (or Azure Cosmos DB)**

Stores scored customer records, campaign history, and feedback labels.

- **Azure Data Factory**

Orchestrates:

- data ingestion,
- feature generation,
- batch scoring pipelines,
- periodic retraining jobs.

- **Azure Key Vault**

Protects secrets, connection strings, and API keys.

- **Azure Monitor / Application Insights**

Monitors:

- latency and availability,
- model endpoint health,
- key metrics for MLOps and business KPIs.

11. Use of Generative AI

For the **core predictive task** (estimating subscription probability based on structured data), traditional supervised models like XGBoost are:

- more efficient,
- easier to control and validate,
- better suited to tabular data,
- more straightforward to govern.

Therefore, **Generative AI is *not* recommended as the primary modeling approach** for the propensity model itself.

However, Generative AI can be highly valuable for:

- **Call script generation** tailored to segments or even individual customers.
- **Summarizing previous interactions** or account history for agents before a call.
- **Drafting personalized follow-up emails or messages** aligned with the call outcome and customer profile.

In summary, Generative AI should be seen as a **complementary technology** that enhances agent productivity and customer communication, rather than as the core predictive engine.

12. Estimated Implementation Timeline

Below is an indicative timeline for delivering an end-to-end **Minimum Viable Product (MVP)**:

1. Week 1 – Discovery & EDA

- Business requirement refinement.
- Data ingestion, cleaning, and EDA.
- Initial target definition and baseline benchmarks.

2. Week 2 – Modeling & Evaluation

- Data preprocessing pipelines.
- Training Logistic Regression, Random Forest, Gradient Boosting, XGBoost.
- Model comparison and selection.
- Initial SHAP explainability analysis.

3. Week 3 – Deployment & Integration

- Packaging XGBoost model and pipeline.
- Azure ML deployment and API exposure.
- Integration with CRM and dialer systems (pilot mode).
- Dashboard creation for KPIs and model monitoring.

4. Week 4 (Optional, Overlapping) – A/B Testing & Governance Setup

- Launch limited A/B test (model-driven vs current strategy).
- Define retraining policy and drift monitoring.
- Complete documentation and handover.

Total Estimated Time:

Approximately **3–4 weeks** for a functional MVP, assuming:

- a dedicated interdisciplinary team, and
- timely access to data, infrastructure, and key stakeholders.

